# Part I

# Finetuning

# Fine-tuning, Reinforcement Learning and Parameter-Efficient Adaptation

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# What we will be talking about

1. Naively Finetuning LLMs
   - What is finetuning?
   - Why not (fully) finetune LLMs?
2. Making finetuning practical through LoRA
   - Understanding why LoRA works
   - How to use LoRA
3. Making finetuning scalable through QLoRA
   - Quantizing LLMs: Why and how?
   - How does it differ from LoRA?

# Outline

Section 1

# Naively Finetuning LLMs

Subsection 1

## What is finetuning?

# Why finetune LLMs?

- **Prompting:** Changes input behavior. We guide the model with instructions and examples.
- **Fine-Tuning:** Changes model parameters. We switch the knowledge of the model.

## Reminder

**Fine-Tuning (FT):** The process of further training a pre-trained model on a smaller, domain-specific dataset to adapt its general knowledge to a specialized task.

# Pretraining vs Finetuning

## Reminder

- Pre-training provides the model with initial, general knowledge by utilizing any and all data sources.
- Finetuning shifts existing knowledge, to specific domains using domain-specific datasets.

# Example: Domain-specific tasks

**Part-of-Speech (POS) Tagging**

The process of assigning grammatical categories to each word in a sentence.

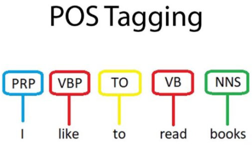



Figure: https://byteiota.com/pos-tagging/

# Example: Domain-specific tasks

## Biomedical POS Tagging

- Patterns absent or rare in general-domain text.
- Prompting probably not enough due to missing knowledge.

| Word | POS Tag |
|---|---|
| Homozygous | JJ |
| null | JJ |
| mutants | NNS |
| die | VBP |
| shortly | IN |
| after | RB |
| birth | NN |

Table: Example from the CRAFT dataset.

Section 1

# Naively Finetuning LLMs

Subsection 2

## Why not (fully) finetune LLMs?

# High Resource Cost

- **Computational Power:** Full FT requires massive hardware requirements and thousands of compute hours for large models (e.g., Llama-70B).
- **Storage:** Each specialized task requires a separate, full copy of the massive foundational model.

# Operational Overhead

- Training is slow.
- Deployment becomes complex, managing multiple large models for different applications.
  - A single 13B model may require 26GB of storage.
  - 10 finetuned models may thus require 260GB.
- High risk of Catastrophic Forgetting—the model forgets its general knowledge while learning the specific task.

# Scalability issues

## BERT

- **Model Size:** ~340M Parameters
- **VRAM Requirement:** ≈ 8-12GB
- **Training Time:** Minutes to a few hours

## Llama 70B

- **Model Size:** ~70 Billion Parameters
- **VRAM Requirement:** ≈1000GB+
- **Training Time:** Days to Weeks (even with specialized hardware)

LLMs are too large to run full-finetuning. But do we need to?

# Parameter-Efficient Fine-Tuning (PEFT)

## Definition

**PEFT**: A set of techniques used to adapt large pre-trained LLMs to specific tasks by updating only a small subset of their parameters, rather than retraining the entire model.

- Many algorithms developed through the years (adapters, prompt/prefix tuning etc.).
- However, the LoRA family of methods has dominated the LLM space in the last few years.

# Outline

Section 2

# Making finetuning practical through LoRA

Subsection 1

## Understanding why LoRA works

# PEFT: LoRA

- Low-Rank (LoRA) Matrix Decomposition is a specific PEFT algorithm that exploits the fact that weight updates in LLMs have a "low intrinsic rank".
- Low rank matrices enable us to scalably represent information.

# Low-rank matrices

## What is a matrix?

A matrix is essentially a organized collection of data (rows and columns). For large datasets, these matrices can become incredibly complex, requiring massive storage and computational power.

## Example: Data matrix

$$\begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 1 & 2 & 3 & 4 & 5 \\ 3 & 6 & 9 & 12 & 15 \\ 4 & 8 & 12 & 16 & 20 \end{bmatrix}$$

# Low-rank matrices

## Matrix rank

- Rank $r$ represents the number of genuinely independent columns (or rows!) within the matrix $M$.
- If $M \in \mathbb{R}^{m \times n}$, then $rank(M) \leq min(m, n)$.

## Example: Music

- Imagine a piano with 88 keys.
- Most songs don't require all 88 keys; they can be described by melody, rhythm etc.
- Low-rank updates capture changing general patterns.

# Low-rank matrices

## Rank Factorization

If a matrix $M$ has a low rank $r$, we can approximate it (or exactly represent it) by factoring it into a product of two simpler matrices.

## Example: Rank 1 matrix

$$\begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 1 & 2 & 3 & 4 & 5 \\ 3 & 6 & 9 & 12 & 15 \\ 4 & 8 & 12 & 16 & 20 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}.$$

# Why are LLM updates typically low-rank?

## Many possible explanations

- Updates generally follow the direction of existing knowledge $\rightarrow$ the updates may often just be re-weighting of existing weights.
- Underlying ML problems are usually inherently low-rank.

# Why are LLM updates typically low-rank?

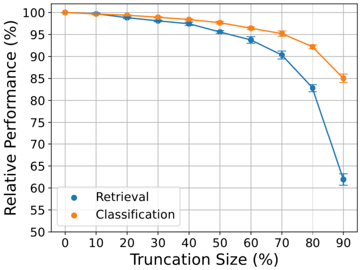



Figure: `https://aclanthology.org/2025.emnlp-main.1410.pdf`

**Large embedding spaces**

Modern NLP models are over-parameterized $\rightarrow$ not enough information to fill in higher ranks of weights.

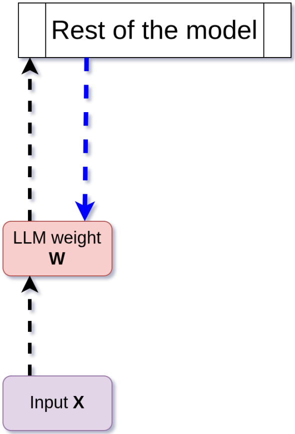
# LoRA: Low-rank decomposition

$$\underbrace{A_{m\times n}}_{\text{large matrix}} \approx \underbrace{U_{m\times r}}_{\text{left factors}} \underbrace{V_{r\times n}}_{\text{right factors}}, \qquad r \ll \min(m,n).$$

$$A_{10000\times 5000} \approx U_{10000\times 50} V_{50\times 5000},$$

$$\underbrace{10000 \times 5000}_{50{,}000{,}000 \text{ parameters}} \longrightarrow \underbrace{10000 \times 50 + 50 \times 5000}_{750{,}000 \text{ parameters}}.$$

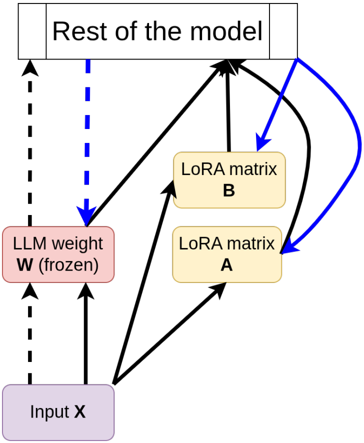
# LoRA: Mathematical Foundation




## Full SFT

- $W \in \mathbb{R}^{d \times k}$ weights.
- $X$ input, $h$ output.
- Fwd pass: $h = WX$.
- Bckwd pass: $W = W + \alpha \nabla h$

# LoRA: Mathematical Foundation



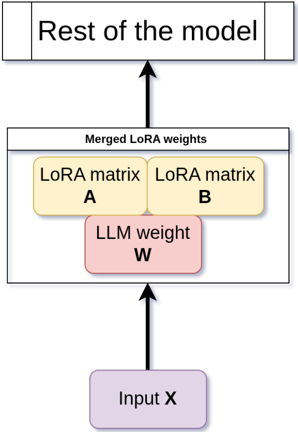

## LoRA

- $A \in \mathbb{R}^{r \times k} \rightarrow$ Gaussian.
- $B \in \mathbb{R}^{d \times r} \rightarrow$ zeros.
- $r \ll \min(d, k)$, hence $A, B$ are much smaller than $W$.

## Example: Music

$A$ could represent the genre of music, while $B$ the melody, rhythm etc.

# LoRA: Mathematical Foundation




## LoRA

- During inference, we compute $H = WX + \Delta WX$.
- Alternatively, merge weights: $W' = W + \frac{\alpha}{r}(AB)$

# LoRA: Algorithm

▷ *====Initialization====* ◁

**for all** $W \in$ *all_weights* **do**

  Set $\mathbf{W}$ to read-only ▷ *Original knowledge preserved*

**for all** $W_u \in$ *target_modules* **do**

  Initialize $\mathbf{A}$ randomly

  Initialize $\mathbf{B} = \mathbf{0}$ ▷ *Contributes zero to the output initially*

▷ *====Backpropagation====* ◁

**for all** $X \in$ *input_embeds* **do**

  $\mathbf{\Delta W} = \frac{\alpha}{r}\mathbf{AB}$

  $\mathbf{H} = \mathbf{W}X + \mathbf{\Delta W}X$ ▷ *Output = original + adaptation*

▷ *====End of training====* ◁

$\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r}(\mathbf{AB})$ ▷ *Merge original and adaptation*

Section 2

# Making finetuning practical through LoRA

Subsection 2

## How to use LoRA

# LoRA: Practical Considerations

- How many ranks do I use?
  - $r \in [1, 16]$: fast, easy, recommended for minor adjustments.
  - $r \in [16, 64]$: recommended for most domain adaptation scenarios.
  - $r \in [64, 256]$: may hit computational bottlenecks, only use for dramatic shifts in domain and knowledge.
- How do I set $\alpha$?
  - Usually set to a multiple of $r$ .
  - Very low $\alpha \to$ minimal contribution. Very high $\alpha \to$ destruction of model knowledge.
  - $\alpha = 2r$ usually used as a starting point.

# LoRA: Practical Considerations

- What modules do I change?
- Start from the top of this list and go down as required:
  1. **Query (q_proj) and Value (v_proj):** Most frequently adapted; control how the model attends to input.
  2. **Key (k_proj):** Increases information retrieval capacity.
  3. **Output (o_proj):** Less common, but useful for modifying integrated attended information.
  4. **Feed-forward Networks (MLP):** Possible to adapt, but generally offer less parameter benefit compared to attention layers.

# LoRA: Practical Considerations

- Should we merge the LoRA weights?

$$\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r}(\mathbf{AB})$$

**Yes**

- Lowers GPU memory and execution time.
  - More significant if $r$ is large.
- Deployment may be easier.

**No**

- You can use the same model with many finetuned adapters which are dynamically selected depending on your task.

# Outline

Section 3

# Making finetuning scalable through QLoRA

Subsection 1

## Quantizing LLMs: Why and how?

# QLoRA: Quantization

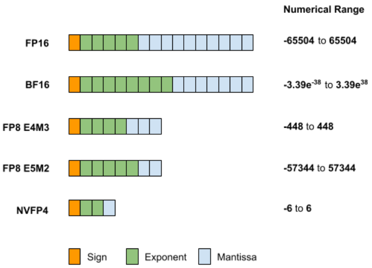



Figure: https://developer.nvidia.com/blog/model-quantization-concepts-methods-and-why-it-matters

- Quantization refers to dropping the size (not number) of the LLM parameters.
- Normally, each parameter is stored in a 32 bit floating point.
- Quantization can drop them to usually 4 bits, resulting in $\times$8 less GPU memory and inference costs.

# QLoRA: Quantization

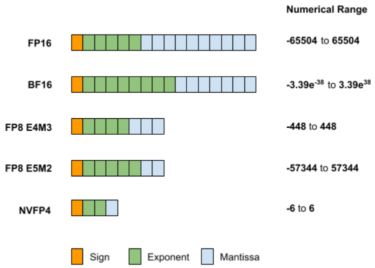



Figure: `https://developer.nvidia.com/blog/model-quantization-concepts-methods-and-why-it-matters`

- Quantization is one of the most useful tools at your disposal.
- Allows for training and inference in consumer-grade hardware.
- We will return to this in Module 5.

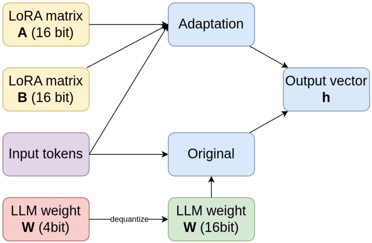
Section 3

# Making finetuning scalable through QLoRA

Subsection 2

## How does it differ from LoRA?

# QLoRA: Concept




- Load a 4-bit quantized model using a normal approximation of weights.
- Run LoRA in full precision, dequantizing the model weights one at a time.
- Continue as normal.

# QLoRA: Algorithm

▷ *====Initialization====* ◁
▷ *Same as before* ◁
▷ *====Backpropagation====* ◁
**for all** $X \in \mathit{input\_embeds}$ **do**
    $\mathbf{\Delta W} = \frac{\alpha}{r}\mathbf{AB}$
    $\mathbf{H} = \mathrm{dequant}(\mathbf{W})\, X + \mathbf{\Delta W} X$ ▷ *Output = original + adaptation*
▷ *====End of training====* ◁
▷ *Only attempt dynamic merge. Ignore this advice at your own peril, or if you really know what you are doing.* ◁

# QLoRA: Pros and cons

## Pros

- Frozen weights no longer monopolize GPU memory and inference time.
- Allows training in consumer-grade hardware.
- Massively reduces training times.

## Cons

- End-result is a quantized $\rightarrow$ less powerful model.
- Dequantization imposes a $\approx 10\%$ tax on training times.
- Merging is now much messier.

# Summing up

- Full finetuning for LLMs is out of the question.
- However, weight updates are usually low-rank.
- Thus, we can approximate full finetuning with LoRA.
- We may use QLoRA instead to make it much more scalable.

# Part II

# Reinforcement Learning

# Outline

Section 4

# General RL

Subsection 1

## Fundamentals

# What is RL?

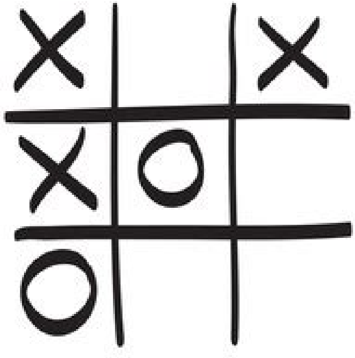
## Definition (Reinforcement Learning)

A machine learning paradigm where an Agent learns optimal behavior through interaction. The system learns autonomously by trial and error, maximizing a cumulative Reward Signal.

## Example

A master chess player makes a move. The choice is informed both anticipating possible replies and counterreplies and by immediate, intuitive judgments of the desirability of particular positions and moves. (Sutton & Bartol: “RL An Introduction”, 2nd Ed).

# Example: Tic-tac-toe



## How would you train an agent?

- Minimax: Thrown off due to suboptimal play.
- Dynamic prog.: Needs a perfect configuration of the opponent.
- A human would learn by doing (and failing).

# Example: Tic-tac-toe

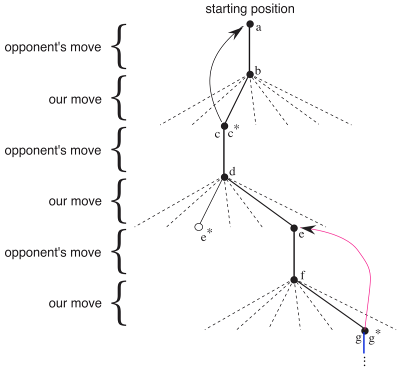



Figure: Sutton & Bartol: “RL An Introduction”, 2nd Ed

## RL

- Set up a table where each cell = one possible ttt configuration.
- All states where no players win are given a value of 0.5 (we do not know who would win).
- All states where we win are 1, otherwise 0.

# Example: Tic-tac-toe

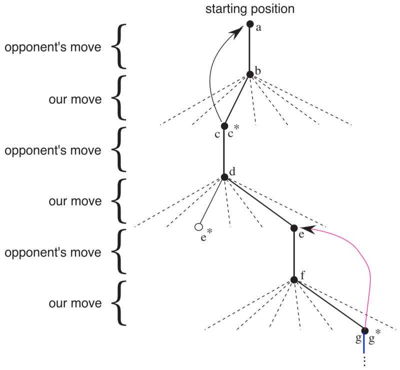



Figure: Sutton & Bartol: “RL An Introduction”, 2nd Ed

## RL

- At each step, consider possible moves.
- Take the best one (exploitation), or a random one (exploration).
- Update previous steps with results. $V[s] \leftarrow V[s] + \alpha[V[s'] - V[s]]$

# Other methods

- In reality, we use much more sophisticated and optimized methods.
- Most nonetheless follow the paradigm of $NewEstimate \leftarrow OldEstimate + StepSize[Target - OldEstimate]$.
- The DPO and GRPO methods used in modern-day LLM training also follow it.
- We obtain those estimates from the environment.

# General method

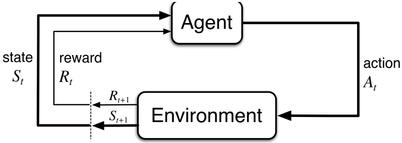



Figure: Sutton & Bartol: "RL An Introduction", 2nd Ed

## The learning loop

At each step (at state $S_t$):

- Take action $A_t$.
- Move from state $S_t \to S_{t+1}$ and observe reward $R_{t+1}$.
- The environment is responsible for providing $S_{t+1}$ and $R_{t+1}$ given $A_t$.

# The Environment

| $s$ | $s'$ | $a$ | $p(s' \mid s, a)$ | $r(s, a, s')$ |
|---|---|---|---|---|
| high | high | search | $\alpha$ | $r_{\texttt{search}}$ |
| high | low | search | $1-\alpha$ | $r_{\texttt{search}}$ |
| low | high | search | $1-\beta$ | $-3$ |
| low | low | search | $\beta$ | $r_{\texttt{search}}$ |
| high | high | wait | 1 | $r_{\texttt{wait}}$ |
| high | low | wait | 0 | $r_{\texttt{wait}}$ |
| low | high | wait | 0 | $r_{\texttt{wait}}$ |
| low | low | wait | 1 | $r_{\texttt{wait}}$ |
| low | high | recharge | 1 | 0 |
| low | low | recharge | 0 | 0. |

Figure: Sutton & Bartol: "RL An Introduction", 2nd Ed

## What is the environment?

All interactions which the agent must not know about.

## Example: Poker

The hands of other players.

## What the environment isn't

If the agent is a paramedic,it should not immediately know the internal injuries of a victim.

# The Environment

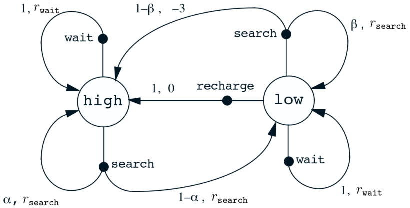



Figure: Sutton & Bartol: “RL An Introduction”, 2nd Ed

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All interactions which the agent must not know about.

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# The Reward

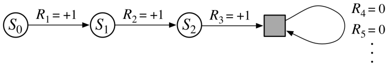


Figure: Sutton & Bartol: “RL An Introduction”, 2nd Ed

## Definition (Reward $r_t$)

A numerical value representing degrees of success given at time $t$.

## Definition (Return $G_t$)

- Discounted returns over $t$ steps.
- $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$.
- As $\gamma \to 1$ the agent prioritizes future rewards.

# Policies and state-value-actions

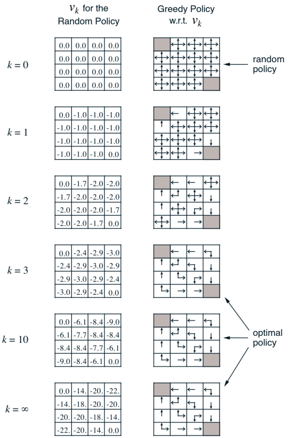


Figure: Sutton & Bartol: "RL An Introduction", 2nd Ed

## Definition (Policy $\pi$)

The agent's strategy, which dictates the behavior by specifying the probability of taking every possible action when the agent finds itself in a specific state.

# Policies and state-value-actions

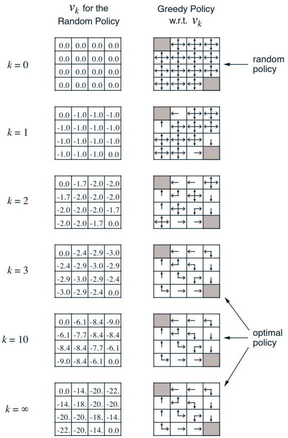


Figure: Sutton & Bartol: "RL An Introduction", 2nd Ed

## Definition (Value State Function $V$)

Estimates the expected future discounted reward an agent will accumulate starting from a particular state, assuming it follows a specific policy.

# Policies and state-value-actions

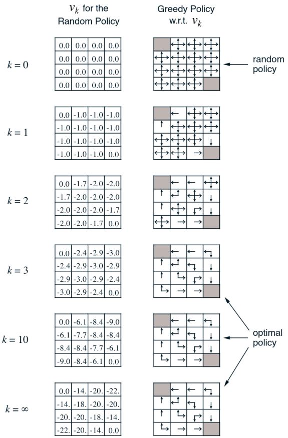


Figure: Sutton & Bartol: “RL An Introduction”, 2nd Ed

## Definition (Value-Action State Function $Q$)

Estimates the expected future discounted reward an agent will accumulate starting from a particular state and immediately taking a specific action.

# Generalizing the RL Loop

Initialize policy $\pi$ and value/Q-function $V/Q$
Initialize Experience Buffer $\mathcal{D} \leftarrow \emptyset$
**while** $D$ not satisfied (e.g., episodes remaining) **do**
  ▷ *Interaction / Data Collection* ◁
  **while** $s$ not terminal **do**
    Select action $\alpha \leftarrow \pi(s)$
    $r, new\_s \leftarrow environment(s, \alpha)$
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s, \alpha, r, new\_s)\}$
    $s \leftarrow new\_s$
  ▷ *Learning / Parameter Update* ◁
  Sample a batch of experiences $\mathcal{B}$ from $\mathcal{D}$
  UpdateParameters($\mathcal{B}$, Algorithm Specific Rules)
  DiscountRewardEstimate $\leftarrow$ Calculate($\mathcal{B}$)

# RL for LLMs

## When training LLMs

We learn a policy $\pi(s|\alpha)$ where:

- $\alpha$ is an action (here next chunk of text).
- $s$ is the state (partial generation process for next token).

Instead of explicit $V$ or $Q$ functions we use an Advantage function as a gradient, which estimates how better an action is given the current state (discounted at time-step $t$ in the future).

# Finetuning

## Pros

- Very easy to implement—continuation of pretraining phase.
- Techniques such as LoRA alleviate drawbacks.
- Model learns to comply with task-specific guidelines, requirements, formats.

## Cons

- Computationally heavy.
- Relies on high-quality, human-curated/created data.
- Overfitting.

# Reinforcement Learning

## Pros

- Not as computationally heavy as SFT.
- Factual accuracy.
- Adaptability.

## Cons

- "State" is evolving text sequence.
- Feedback is sparse, delayed and subjective.
- Multiple conflicting objectives.

RL has greater potential, but requires careful planning, since LLMs are not ideal targets for RL.